

Ignis: Eulerian Fluid Simulation and Rendering at VR Frame Rates

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Figure 1: From left to right: (a) Two smoke rings collide. (b) A spinning dust devil. (c) A wall of fire, showing a comparison between dithering and naive undersampling. All views captured in headset on a Valve Index, simulated and rendered on an NVIDIA RTX 3080 at 90 FPS or more with Ignis.

ABSTRACT

We present Ignis, a GPU Eulerian fluid solver capable of simulating and rendering at VR resolutions and refresh rates. Core to our approach are an approximate shadowing technique and an adaptive dithering technique, which allow us to cheaply render fully lit volumes at high resolutions and under a range of visibility conditions. We discuss the design decisions which enable our solver to run at interactive rates.

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1 MOTIVATION

Pre-rendered volumetric effects are common in games and VR simulators in the form of camera-facing flipbooks. However such flipbooks are not interactive, and their illusion breaks down as they get closer to the camera. Use cases which require proximity and interactivity, such as VR training simulators for fire extinguishers /

fire fighting, would benefit from an interactive VR simulation of smoke, fire, and other volumetrics.

Our requirements for runtime fidelity are much higher than a flipbook. While a flipbook may run at a reduced framerate, an interactive simulation must run at the refresh rate of the headset to avoid input lag. Second, while flipbooks are typically rendered as effects in the distance, in VR our user is immersed in the simulation volume, with single voxels occupying an unbounded amount of screenspace. So, we must be able to perform expensive raymarch operations on every pixel in each of the HMD's high resolution displays. These challenges, and others, motivate the design of Ignis' volumetric solver and renderer, which we discuss in the remainder of this paper.

2 SIMULATION

Ignis utilises a dense Eulerian fluid solver using operator splitting [Stam 1999] to approximately solve the Navier-Stokes equations. We track, at each grid cell centre, a velocity, density, and temperature which are each stored in 3D textures with 16-bit float formats. Using half-precision storage has negligible impact on the visual quality of our simulations, while effectively doubling their performance on bandwidth constrained portions of the solver.

Each simulation presented is specified using a custom injection kernel which hooks into the rest of our solver. These kernels receive as input the voxel position, and timestep, and add density and temperature to the simulation, as well as applying arbitrary forces, such as buoyancy.

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Table 1: Average frametimes in milliseconds for solver stages and total simulation & render

Scene	Grid Size	Injection	Advection	Vorticity	Projection	Shadows	Raymarch	Total
Smoke Rings	232 x 232 x 160	0.28ms	2.92ms	N/A	1.91ms	0.82ms	3.69ms	9.63ms
Dust Devil	160 x 256 x 160	0.21ms	2.29ms	0.32ms	1.49ms	0.57ms	4.77ms	9.65ms
Wall of Fire	200 x 200 x 200	0.26ms	2.79ms	0.38ms	1.80ms	0.67ms	3.82ms	9.73ms

2.1 Advection

We use semi-lagrangian advection with a MacCormack backtrace [Selle et al. 2008]. Though we use linear sampling for velocity during the backtrace, cubic interpolation is used for our final resample of all advected properties, and we also optionally apply a vorticity confinement force as in [Fedkiw et al. 2001].

2.2 Projection

Pressure projection is solved using geometric multigrid, using a single V-cycle with 1 pre and 2 post-smooths, and the fullweight and trilinear transfer operators described in [Trottenberg et al. 2000]. Jacobi iteration is used for smoothing, and we approximately solve on our 3rd, coarsest, grid using red-black Gauss-Seidel with successive over-relaxation (SOR).

3 RENDERING

Our volumetric renderer supports up to two shadowed directional lights, anisotropic scattering using the Henyey-Greenstein phase function, and light emission due to blackbody radiation. In VR, pixel counts can easily exceed voxel counts, so we opt to move as much of the rendering process onto the simulation grid as we can.

3.1 Approximate Shadowing

Rather than performing a full shadow raycast per voxel (or worse, per primary raymarch step), we utilise an approximate shadowing algorithm. Baking axis-aligned light extinction into two RGB textures is commonly used to relight VFX flipbooks in real-time use cases, a technique typically referred to as 6-point, or 6-way lighting. Extinction in arbitrary light directions is then approximated by linear combination of the extinction along the appropriate axes.

We extend this technique to work on an interactive simulation by observing that axis-aligned extinction can be computed via cumulative products of the extinction along each axis. In practice since each axis writes to a different texture, we can schedule these compute kernels to run concurrently.

Since our primary raycast is mainly bottlenecked on GPU bandwidth, we also opt to collapse the axis aligned extinctions down to two interpolated directional extinction values. This allows us to compute approximate shadows along our raymarch without the full bandwidth cost of accessing all 6 directional shadow textures.

Even with this additional packing step, precomputation of shadows costs <1ms on all of our experiments.

3.2 Adaptive Dithering

Despite using a packed shadow texture, we cannot afford to exhaustively sample every voxel along our primary raymarch at VR resolutions. Naively undersampling the volume however will produce distracting slicing artifacts. Previous work on real-time volumetric

lighting [Gjoel and Svendsen 2016] utilises ray dithering to break up banding, and this technique also works for primary visibility. However, these approaches rely on TAA to clean up the resulting noise, which in VR produces unsightly smearing artifacts due to the constant movement of the HMD.

We instead opt to use dithering without any temporal accumulation. In practice, due to our high target resolutions, differing dithering patterns in each eye, and VR driver applied distortion, the resulting image is crisp, temporally stable, and renders quickly. We can even adapt our step size to varying performance conditions, which we refer to as *adaptive dithering*. Adaptive dithering is critical to meeting our frame targets, particularly as shading costs vary. For the scenarios in our supplementary video, adapted step counts vary by a factor of 4. Without dithering, as shown in the video, this results in juddering of the image. With dithering we have a stable image free of artifacts except for a small amount of noise.

4 RESULTS

As shown in Table 1, we tested our solver on a range of scenarios designed to test quality and performance under varying densities, velocities, and light levels. All simulations were run with a 1.5cm voxel scale, and rendered at a resolution of 2016 x 2240 x 2. The solver is implemented in DirectX 12 and OpenVR, using a NVIDIA RTX 3080 and a Valve Index HMD. In all scenarios our frametimes are well within the 11.1ms frame budget for a 90 Hz HMD.

5 LIMITATIONS AND FUTURE WORK

Currently, Ignis is primarily limited by the low simulation resolution necessary to run in VR on current hardware. However, many games and training use cases may only require full interactivity on a portion of a volumetric scene. Future work will investigate the composition of interactive and pre-simulated volumetrics in order to enable VR volumetrics with larger bounding volumes and voxel resolutions. In addition, Ignis will be extended to simulate thin flame combustion, and respond to collisions with obstacles in the environment.

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